# Improved Wireless Image Sensor Network to Monitor Tiger Movement

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***Abstract*—Over the past decade Wireless Sensor Networks has been vigorously researched, the nature of Wireless Sensor Networks has made it an ideal application for wildlife monitoring. TigerCENSE was one of the attempts to use wireless sensor network to conserve tigers, one of the most endangered species in the planet. TigerCENSE was an attempt to monitor movements of Tigers using passive infrared sensor, CMOS image sensor and Infrared Flash based wireless image sensing nodes, with ability to wirelessly transmit images taken. We improve upon the existing TigerCENSE node by using emerging standards of 6LoWPAN or IPv6 for Wireless Sensor Network for communication, RPL as a routing protocol, Atmega128RFA1 as processing and communication unit and by using ContikiOS as operating system for our Wireless Sensor Nodes.**

## I. Introduction

Rapid development in MEMS based technology and minaturization has led to emerging field of Wireless Sensor Network which combines sensors, processing unit in form of low powered micro-controller and communication unit in form of Radio transceiver into a single node. These nodes are designed often to operate on a pair of battery for time-span of years and often operate in extremely harsh environments, for example Wireless Sensor Network were deployed in extremely harsh and difficult environment of Active Volcano for Seismic and Infra-sonic signals measurement[1]. The nature of these nodes have made these an ideal application for wildlife monitoring and many projects have tried to leverage this to improve traditional methods of wild life monitoring, one of the first such project, was famous Great Duck Island Experiment[2]. There have been similar projects proposed suited to conditions and nature of wildlife in India,these projects have tried to use concept of Wireless Sensor Network to help in monitoring wildlife, WildCENSE a GPS based animal tracking sensor network was one such attempt[3], TigerCENSE was another such attempt to improve upon existing camera trap technology[4].

Images are important tool for monitoring wild life and rapid advances in technology have led to easy and widespread availability of low cost camera modules. Traditionally, for wild life monitoring camera traps have been used extensively, these camera traps used existing commercially available cameras and motion detectors in form of pressure pads or infrared beam being obstructed to trigger connected camera and take images. These were clumsy, bulky and did not facilitate wildlife researchers to monitor in real-time collected data as these camera-traps lacked capability to wirelessly transmit collected images. Further, the latency between the images snapped was of the order of minutes, this prevented many fast moving animals from being captured. These camera-traps were usually costly, and often caused disturbance to animals as they produced clicking sound while taking images, as these camera-traps used commercially available off-the shelf camera not designed specifically for purpose of wild-life monitoring. Finally, lack of Wireless capability also meant that it was impossible to determine health-status of the node without physical inspection, this was time-consuming and often it was difficult to visit these nodes owing to difficult topography of the forests and reserves. Camera traps have been widely used in monitoring tiger trails, TigerCENSE was an attempt to overcome some of these limitations of camera traps.

TigerCENSE as discussed in [4], was a Wireless image sensing node designed for replacing camera-traps specifically for purpose of monitoring tigers, length and breadth of TigerCENSE node was similar to that of credit card, it came with a radio transceiver that allowed for Wireless communication and transferring of images from the node along with health parameters of the node and came with PIR or Passive Infrared sensor for detecting movement near the sensor node, it was equipped with VGA camera to take images, further to allow for illumination at night without disturbing the animal and take images in dark, it came with an Infrared flash.

There were some drawbacks and problems with the earlier version of TigerCENSE. The previous version of TigerCENSE had custom networking protocol written over Layer 2 Radio XBee 802.15.4[5], this did not have support for multi-hop communication, multi-hop is more energy efficient and increases range of communication. Additionally, the quality of images obtained were also not satisfactory. The entire code base of TigerCENSE was written in C and was close to 4-5,000 Lines of Code, it was complex and difficult to maintain. It did not take advantage of some of the emerging software abstractions for Wireless Sensor Network like standardized protocol stack, programming constructs like Protothreads or using operating system like ContikiOS. Finally, over the past couple of years many chips integrating radio transceiver with

micro-controller have come out, which offer exceptional performance, are extremely small in package, are energy efficient, and thus form an excellent replacement for some of the components used in earlier design of TigerCENSE.

The Research Contribution of this paper is in proposing design of Wireless Image Sensing node specifically designed for improving upon camera-traps for monitoring movement of Tigers through tiger trails. In this paper we discuss the hardware of the proposed wireless image sensing node, improvements made over existing TigerCENSE node in terms of hardware and software. We further discuss how software abstraction offered by ContikiOS in particular Protothreads, helped us in the design of the node. We further discuss networking of these nodes using 6LoWPAN or IPv6 for Low Powered Wireless Sensor Network and evaluation of performance of these nodes as tested in laboratory.

In this paper, Section II deals with discussion on related Wireless Image Sensing platforms, discussion on our prior attempts in improving TigerCENSE node and reasons for choosing 6LoWPAN to network these nodes. Section III deals with the System design of the proposed node in terms of hardware and software, Section IV deals with evaluation of performance of the node in terms of delays and energy consumed and experimental result obtained from test conducted in the laboratory, finally Section V concludes this paper.

## II. Prior and Related Work

Over the past couple of years, a number of Wireless Image Sensing platforms have been proposed, CMUcam[6], CITRIC[7], Cyclops[8] being some of them. These are general purpose nodes that have been used for varied experiments. To the best of our knowledge, TigerCENSE was one of the first Wireless Image Sensing node that was developed specifically for purpose of monitoring tiger movements and overcoming limitations of traditional camera traps. TigerCENSE was unique in many ways, at its core was extremely resource constrained 8-bit Atmel Atmega1281 micro-controller, with a SRAM of 8 Kilobytes, this imposed additional challenges in designing system. Most of the Wireless Image Sensing platform have been based on more powerful 32 bit micro-controller, with RAM and ROM even ranging to scale of Megabytes. Cyclops was exception to this and was based on 8-bit Atmega128L micro-controller, similar to TigerCENSE, however, it used element of CPLD to do processing.

We have progressively developed new version of TigerCENSE, Initially, we tried to optimize existing node, however, it was found more prudent to develop a newer version of the node taking advantage of modules like Atmel Zigbit integrating both Radio Transciever and Micro-controller in a single unit and standardized protocol stack, Zigbee and subsequently 6LoWPAN.

We developed an initial version of node using Atmel Zigbit[9] and Bitcloud Embedded Software Stack[10]. Atmel Zigbit combines Atmega1281 microcontroller[11] and

Fig. 1. Custom PCB of Atmel Zigbit based design of TigerCENSE node

AT86RF230 radio transciever[12] in a single module, this had advantage of reducing cost, size, improving performance. We developed initial version using Meshnetics Meshbean kit[13], we further developed our custom PCB for Zigbit based version of TigerCENSE node, as shown in Figure 1. However, we encountered issues because on-board radio transciever AT86RF230 shared same SPI bus with micro-controller, which was used for interfacing with MicroSD card, additionally, Zigbee implementation in Bitcloud was closed source preventing us from optimizing it according to our application needs. We subsequently developed new node using Atmega128RFA1 and using ContikiOS[14] and 6LoWPAN or IPv6 for Low Powered Wireless Sensor Network to network these nodes, Which we have discussed in this paper.

There has been an increased focus towards using existing standards of IPv6 for sensor network in the past few years, as the sensor networks have become ubiquitous, consequently IETF charted 6LoWPAN working group for adaption of IPv6 for networks using IEEE 802.15.4 PHY layer and defines how IPv6 packets are carried over IEEE 802.15.4 links. The use of 6LoWPAN in sensor networks allows for seamless connectivity of these sensor network to Internet, which otherwise needed Application gateway. The availability of implementation of 6LoWPAN in popular operating system ContikiOS for Sensor Network, the advantage of being totally open standard, and easing connectivity of sensor nodes to Internet motivated us to use implementation of 6LoWPAN as provided in ContikiOS to network nodes. ContikiOS further provided programming abstraction Protothreads that helped in structuring code.

## III. System Design

The design of TigerCENSE system could be divided into hardware architecture of the node and the software that interfaces different sensors and peripherals and handles communication. We use commercially off the shelf available electronic components to develop TigerCENSE node, we developed node progressively moving from Atmel Zigbit based design to Atmel Atmega128RFA1 based design of the node. Figure 2, gives an indicative block diagram based description of hardware architecture of TigerCENSE node.

In sub-section A, we discuss hardware design of the sensor

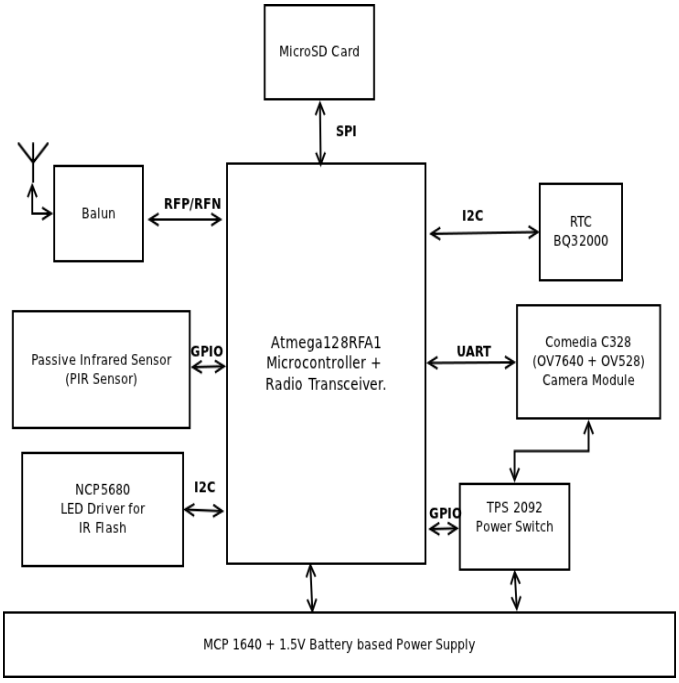


Fig. 2. Block diagram based Architectural overview of the system depicting different components and their interfacing with micro-controller

network node and further we discuss improvements over the earlier design of the node and reasoning behind making these changes in the hardware. In sub-section B, we discuss software of the node in particular, we discuss software abstractions offered by ContikiOS specifcally Protothreads and how it helped us in structuring code that interfaced with Camera, we further discuss issues related to networking of these nodes.

### A. Hardware Design

*1) Microcontroller:* Atmega128RFA1 is a chip combining microcontroller and 2.4GHz low power radio transciever, temperature sensor into a single unit. It is derived from Atmega1281 microcontroller[11] and is designed for Wireless Sensor Network applications in mind. The backward compatibility of the modules with Atmel Atmega1281 allowed for significant code reuse and allowed us to use interfacing code of some of the peripherals from the earlier design of the node. Atmega128RFA1 includes Low power radio transciever similar to Atmel AT86RF231[15] within the chip and thus frees the SPI bus, unlike Atmel Zigbit modules[9]. Atmel Atmega128RFA1 includes on-board temperature sensor, 128 Kilobytes of Program Flash Memory, 16 Kilobytes of SRAM, 4 Kilobytes of EEPROM. Atmega128RFA1 comes in 64 pad QFN packaging which makes it extremely compact, this allows for us to reduce size of board significantly, reducing size is useful as this allows us to conceal these nodes in environment easily thus reducing probability of theft of these modules. Atmega128RFA1 does not have integrated filter balun or chip antenna and these has to be connected externally. We decided to use 2.45Ghz filter balun 2450FB15L0001E[16] and 2450AT45A100E 2.45Ghz chip antenna[17] for our application. We have programmed fusebits of microcontroller to operate at clock frequency of 8MHz.

The initial prototype of TigerCENSE node was designed using deRFmega128 modules[18] on custom board we have designed in laboratory, these modules combine Atmega128RFA1 chip with chip-antenna, balun in a single module, thus simplifying design at the initial stage of project.

*2) Radio Transciever:* Low-Power radio transceiver in Atmega128RFA1 operates at frequency range of 2.4 Ghz which is ISM band, the radio transceiver has Link Budget of 103.5 dB and receiver sensitivity of -100dBm and programmable output power ranging from -17dBm to +3.5dBm. Further, it has ultra low-current consumption of 12.5mA in receive mode and 14.5mA in transmit mode at maximum transmit power. Though, the most promising feature is support for PSDU rate as high as 2 Megabits/second. The higher data rate becomes extremely useful for our application, where amount of data communicated is large.

*3) Real Time Clock :* Real Time Clock (RTC) is used to maintain time on node, this is important to record time of happening of the event. RTC is synchronized to time at the base-station. We use Texas Instruments BQ32000 RTC chipset[19] in our node, this needs an external crystal with resonant frequency of 32.768 kHz and operates at 3-3.6 volts. It works over $I^2C$ interface and typically consumes 100 µA of current when active, in the active mode we can communicate with RTC using $I^2C$ bus. RTC consumes 1.2 µA of current in backup supply voltage mode which only maintains time. RTC is kept in this low-powered state most of the time, RTC is switched to Active mode only at time of storing of images in MicroSD card. The information about time of taking image is stored along with the file, the headers describing the file in our custom file-system has additional field to record time when the file was created, at the time of sending image to the base-station we also send information about when the image was taken.

*4) Power Distribution :* In our nodes all the sensors and components can operate at voltage of 3.3 Volts, as the node is battery powered, with time, the battery voltage reduces and thus we used buck-boost DC-DC converter to regulate DC voltage, in the earlier node we used TPS63001 buck-boost DC/DC converter[20]. We have replaced TPS 63001 with MCP1640 dc-dc step up converter[21]. This allows us to drive entire node on a single AA/AAA 1.5 Volts pencil battery. MCP1640 is 96% efficient and has a low startup voltage of 0.65 Volts and can drive current $> 350mA$ at output voltage of 3.3 Volts and Input Voltage of 2.4 Volts. We plan to power the node of TigerCENSE on a single AA battery, this would give output voltage of 1.5 Volts, MCP1640 gives typical output current of 150 mA at 3.3 volts at 1.5 Volts input, even if voltage drops to 1.2 Volts, MCP1640 should be able to drive $> 100mA$ , sufficient to drive all the peripherals on board.

We use TPS2092 power-switch[22] to turn off camera module when not in use, we also use this to turn off RTC power supply and put into backup-power supply mode, which consumes less current and only maintains time.

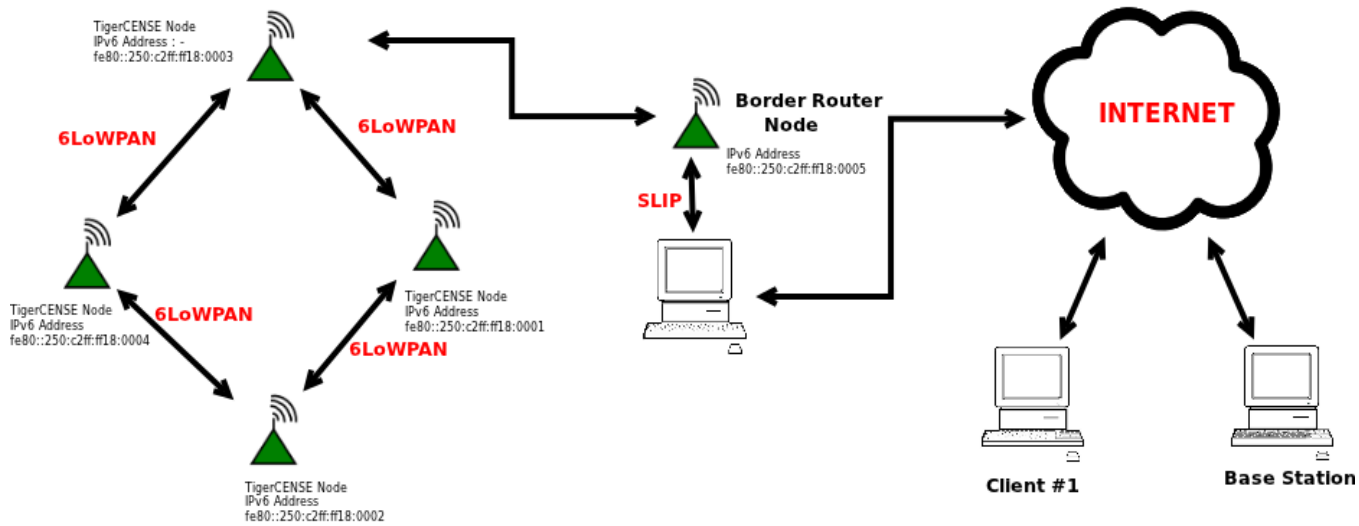


Fig. 3. Overview of TigerCENSE system with individual nodes having IPv6 address,border-router bridging network of 6LoWPAN nodes with Internet and base-station located remotely.

*5) Image Sensor:* We are using Comedia C328 Camera module[23] for our Wireless Image Sensing node, this remains unchanged from the earlier design of the node. Comedia C328 includes omni-vision OV7640 image sensor, this is connected to JPEG compressor chip Omnivision OV528, the compressor chip gives us interface in form of UART to interface and subsequently control camera module and fetch images. The module takes VGA resolution images, these images are compressed and are in JPEG format. The camera module draws 60mA of current at 3.3 Volts, when it is operating and in low-powered inactive state it draws 100 µA of current. The camera module does not have IR cut filter, thus allowing us to take images under Infra-red flash.We also tested Comedia C628 camera module[24], however, because of presence of IR cut filter in lens, lack of low power mode and packet-size of 64 Kilobytes making it impossible to keep single packet in SRAM of Microcontroller, we reverted back to Comedia C328 camera module.

*6) Storage:* The use of Image sensors has resulted in these nodes generating huge amount of data that needs to be stored and communicated, each of these introduces new challenges in implementing system. On an average we found image generated by our image sensor to be of resolution 640x480 pixels and around 40 Kilobytes in size. This cannot be stored in onboard memory of microcontroller, additionally, wildlife researchers might want to fetch these images at a later moment of time from the node instead of communicating these images as soon as they are snapped, which needs signficant amount of local storage. In our node we have interfaced MicroSD card to the microcontroller through SPI bus operating at 2 MHz. MicroSD card offer capacity in range of Gigabytes, which is enough for storing thousands of images.

*7) Passive Infrared Sensor:* PIR Sensor is used to record movement near the node, we are using PIR sensor from Parallax numbered 555-28027[25]. This works at 3.3-5 Volts and draws 100 µA of current. PIR sensor has Fresnel lens with Field of View angle of 90 degrees and range of approximately 20 feet. The PIR sensor at the beginning waits for time of 10-60 second to form heat-map of the surrounding, and interrupts from PIR sensors are disabled during this time period. If there is an object emitting infraread radiations, like living beings,in the FOV of PIR sensor, it disturbs the heat-map generated by PIR sensor and subsequently PIR sensor raises interrupt to the microcontroller, microcontrolle is in energy conserving low-power sleep state, on receiving interrupt wakes up, initializes camera, triggers Infrared-Flash and takes Image and stores it in local storage.

*8) Infrared Flash :* IR LED need to be chosen in a way that it illuminates the entire viewing angle of the camera module. We are presently using earlier design of Infrared Flash as described in [4]. We have tested our existing node using the older IR Flash design.

We are also developing a NCP5680[26] based circuit to control the IR LED, this should improve significantly the quality of images obtained. NCP5680 allows one to control IR LED(s) independently using $I^2C$ interface, it has features like Auto cutoff which is very useful for the application to get better quality images. The IR Flash circuit we are designing based on NCP5680, using a super capacitor should be able to generate enough current to drive Infrared LED for time as large as 100 ms.

### B. Software Architecture

*1) Operating System:* Contiki is a research driven operating system for Wireless Sensor Network and is modular in design and comes with different software abstractions and protocols to simplify programming resource constrained wireless sensor nodes [14]. uIP stack was one of the first attempts to demonstrate that it was possible to have TCP/IP stack running on constrained environment of microcontroller which form heart of Wireless Sensor nodes. [27]. uIP has been further

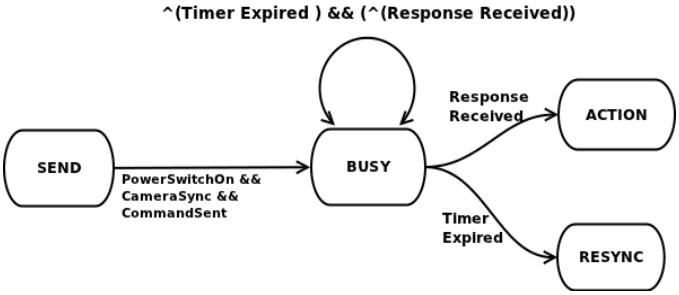


Fig. 4. State Machine representation of typical command sent to camera module

developed to support IPv6[28], uIPv6 is included with the ContikiOS and provides IPv6 based networking between sensor nodes. ContikiOS also includes support for IETF candidate for IPv6 routing in low-power wireless network, RPL routing protocol[29]. ContikiOS further provides us with software-abstraction in form of Protothreads, that makes programming simpler by providing linear-code execution[30]. We use these features of ContikiOS in TigerCENSE, we use protothreads extensively in interfacing of camera module, we further use uIPv6 to network these nodes and RPL routing protocol for making routing decisions.

*2) Protothreads:* At the heart of Wireless Sensor Nodes is extremely resource constrained 8-bit microcontroller, these are constrained by memory and number of CPU cycles they can execute per second. In such a constrained environment, event-driven programming is commonly used, a thread based system usually requires memory to be allocated for stack for each thread and thus increases memory consumption substantially. This is the reason most platforms for constrained wireless sensor nodes like TinyOS[31] or Embedded Software Stack like Atmel Bitcloud use event-driven model of programming. These event-driven model introduce significant problems in introducing blocking wait and often involves writing separate states to implement this. The intermittent version of TigerCENSE which was based on Atmel Bitcloud was similarly structured as one big state machine had similar problem. Protothreads is a programming abstraction that simplifies programming complex event-driven programs by providing conditional blocking statement [30]. Protothreads is included with ContikiOS, and based on our experience of writing TigerCENSE code in totally event-driven model of Atmel Bitcloud and using Protothreads in ContikiOS, it simplified programming substantially.

The camera module Comedia C328, used in our node supports 10 commands, these are sent through UART port of the microcontroller. Looking at implementation of commands sent to camera module, we have identified following structure in command sent to camera module. In an event driven, state machine, based implementation of the command, the state machine looks like as illustrated in Figure 4. An implementation of the state machine in pseudo code is shown in Listing 1.

```
state: { SEND, BUSY, ACTION, RESYNC}

CameraCommand:
        if( state := SEND )
                if(PowerSwitchOn AND CameraSync)
                        if( SendCommand )
                                StartTimer();
                                state = BUSY;
                else
                        return FALSE;
        else if ( state := BUSY )
                if(CameraResponse AND !TimerExpired)
                        state = ACTION;
                else if ( TimerExpired )
                        state = RESYNC;
        else if (state := ACTION )
                // Perform some action
                // corresponding to command sent
        else if ( state := RESYNC )
                ReSyncCamera();
                return FALSE;
```

Listing 1. Pseudocode representation of state-machine to send command to Camera Module.

The code in Listing 1 was simplified considerably using Protothreads, equivalent code of Listing 1 using Protothreads is shown in Listing 2. The number of lines of code is reduced, further, This replicated for all the commands of the camera module, leads to simplified code for interfacing of camera module.

```
PthreadCameraCommand:
        PT_BEGIN
        if(PowerSwitchOn AND CameraSync )
                if( SendCommand )
                        StartTimer();
        else
                return FALSE;
        PT_WAIT_UNTIL(!TimerExpired OR CameraResponse);
        if( CameraResponse )
                // Do Some Action
                return
        ReSyncCamera();
        return FALSE;
        PT_END
```

Listing 2. Pseudocode representation of code in Listing 1 using Protothreads

*3) File System:* Presently we have written an extremely simple file-system for storing files. We plan on using Coffee File System instead. Coffee File System for ContikiOS has been designed considering characteristic of Sensor Network in mind [32]. We are presently working on porting Coffee File System to our platform, porting involves writing procedures that interfaces to MicroSD card and provides ability to write a sector and read a sector, these procedures would be then used by CFS to maintain File System.

*4) Border Router:* Border Router is a computer that serves to connect two networks, network of 6LoWPAN TigerCENSE nodes with Internet, This is similar to Application gateways used traditonally in Wireless Sensor Network which do not use 6LoWPAN protocol stack, however, unlike Application Gateways, the border router does not have any specialized application running, or it does not need to understand different protocols for instance Zigbee. Border Router computer is connected to one of the TigerCENSE nodes running Border-Router application of ContikiOS, it is responsible for encapsu-

TABLE I
ENERGY CONSUMPTION ESTIMATION OF TIGERCENSE

| Sr.No | Component | Mode | Current Consumption(mA) | Time of Operation(s) | Energy Consumed (mA-h) |
|---|---|---|---|---|---|
| 1 | C328/Camera | Active | 60 | 89.6 | 1.493 |
| 2 | MCP1640/DC-DC Booster | Active | 0.019 | 86400 | 0.456 |
| 3 | BQ32000/RTC | Active | 0.1 | 65.6 | 0.0018 |
| | | Low-powered | 0.0012 | 86334.4 | 0.0288 |
| 4 | PIR Sensor | Active | 0.1 | 86400 | 2.4 |
| 5 | Microcontroller | Active(RX/TX) | 18 | 80 | 0.4 |
| | and Radio Transciever | Active (Radio Idle) | 6.5 | 953.6 | 1.721 |
| | | Sleep | 0.001 | 85366.4 | 0.0237 |
| 6 | MicroSD card | Read | 50 | 5.92 | 0.082 |
| | | Write | 50 | 8 | 0.11 |
| | | Initialize | 60 | 1 | 0.017 |
| 7 | 12 IR LED(s) based Flash | ON | 12000 | 0.4 | 1.333 |

lating and transmitting IP packets from Internet to 6LoWPAN network of TigerCENSE node and vice versa, microcontroller in border-router node communicates with the computer using SLIP protocol.

*5) Base Station:* The base station is responsible for maintaining repository of images taken from different nodes, the base-station unlike in previous version of TigerCENSE could be located anywhere in the world as long as it has a IP address which network of TigerCENSE nodes can open socket with and connect to. Presently, we communicate between TigerCENSE nodes and the base station by establishing a TCP connection between the TigerCENSE node and the Base-station.

Having ability to connect directly to individual nodes, allows for possibility of not having intermediary storage of images at the base-station, however, maintaining repository at base-station is necessary as there is a possibility that when we want to retrieve image from individual node, it might be in low-powered sleeping state to conserve energy or might have run out of battery or in worst-case could even be stolen.

*6) Routing :* Contiki provides an implementation of RPL protocol, which is IETF candidate protocol for Routing on Low Powered Wireless Sensor Network. ContikiRPL is included with ContikiOS and has to be enabled for its support to be included[29] . We are using ContikiRPL for routing over 6LoWPAN links, this allows for multi-hop communication with different nodes in the network, which was one of the problems in the earlier implementation of TigerCENSE node.

## IV. PERFORMANCE EVALUATION AND ENERGY ESTIMATION

### A. Goodput

To measure maximum possible goodput of the node, we created simple application, the server application opens up a UDP socket and waits for data. At the client side we use Event Timers provided by ContikiOS to periodically request ContikiOS to send UDP data to server(s) IP address. This is scheduled at every Tick, In ContikiOS, 1 second has 128 ticks. At server, we fire Event Timer every 1 second and calculate number of packets and size of application payload received, from this we calculate goodput.

We found that as the payload of data sent increases the goodput increases. We measured goodput for different payload size, the average goodput achieved for different payload size is shown in Table II. The measurements were made keeping nodes (approx) 50m apart. Goodput was calculated every second at server and outputted through UART, this was done for 3 minutes and average goodput was calculated from all the collected goodput values.

TABLE II
GOODPUT VS PAYLOAD

| Application Payload(bytes) | Average Goodput (bytes/second) |
|---|---|
| 50 | 3189 |
| 100 | 6379 |
| 150 | 6290 |
| 200 | 8321 |
| 250 | 10636 |

We find, for a Payload of size of 250 bytes we found goodput to be about 10.39 Kilobytes per second.

### B. Delays

To evaluate the latency involved in reading and writing to Memory Card, we toggled a GPIO pin of microcontroller before sending command to MicroSD card and toggled it back after function returned. This was captured on an oscilloscope. We found it took 50ms to initialize the MicroSD card, further it took 5ms to write 512 bytes to the MicroSD card and approximately 3.7ms to read 512 bytes of MicroSD card. MicroSD card was interfaced using SPI bus operating at clock frequency of 2 MHz.

In our node UART based image sensing module is connected to microcontroller and can operate at maximum baud rate of 115200 bits/second. A typical image snapped by this module varies in size between 15-40 Kilobytes. The UART based camera module allows us to retrieve images in discrete packets with maximum size of 512 bytes, at UART transfer rate of 115200 bits/second (ignoring overhead of start,stop and parity bits), it would take approximately : $(512 * 8)/115200 = 36ms$ to fetch one packet from camera module, further, for MicroSD card operating at SPI clock of 2 MHz we calculated it takes approximately $5ms$ to write to memory card. Thus, fetching packet from camera module and writing single packet would take approximately $41ms$, if we take size of image to be $40Kilobytes$, we would need to fetch 80 packets from the camera module, this would take total time of $80 * 41ms = 3.28seconds$, during this time we disable the interrupts, to avoid existing data from being corrupted,this introduces delay between two subsequent images that can be snapped.

### C. Energy Estimation

We try to estimate energy consumption of node for a single day, we assume there are 20 events recorded on a single day. If the total time of operation is $86400seconds$ we find that total energy consumption of node over period of 24-hours is 8.0663 mA-h. Refer Table I.

MCP1640 DC-DC converter is 96% efficient, a single AAA Duracell battery(MN2400LR03) provides 1175 mAh energy, of this 96% i.e 1128mAh is usable. This means that we would need to change battery every $1128/8.0663 = 140days$.

To calculate energy consumption, we have assumed PIR sensor and DC-DC booster to be always active because of inherent nature of these components, additionally the following assumptions regarding different components have been made to estimate energy consumption.

*1) Camera :* In each command sent to camera, there is worst case timer event of 300ms within which camera must respond, thus, the worst case time to setup camera will be $4 * 300ms = 1.2seconds$. We have already discussed it takes $3.28seconds$ to fetch image from camera module and store it in MicroSD card. We manually switch off camera through TPS 2092 switch to prevent energy consumption due to quiescent current of camera. In a single day we would be operating camera for duration $(20 * (3.28 + 1.2)) = 89.6s$.

*2) RTC :* Real Time Clock remains active on 100 μA only at time of storing image, when micro-controller needs to find time of taking snapshot. The rest of the time it is kept in Backup power supply mode, which only maintains time and consumes 1.2 μA of current.

*3) Microcontroller and Radio-Transciever:* We assume that suitable MAC layer protocol implementing radio duty cycling mechanism is used to keep radio switched-off for majority of time to prevent Idle listening and conserve energy. We assume that 99% of time radio is powered off and 1% of time it is active, during the time it is active, it is either Idle listening the channel for communication, transferring images to base-station or communicating with other nodes to maintain routing table. We collect 10 images and store them in MicroSD card and subsequently send them to the base-station in one batch. We assume base-station to be located at a single-hop from the TigerCENSE node, and goodput to be 10 Kilobytes/second based on some of the readings we have taken earlier.See table II. We acknowledge over multi-hop link, goodput will be significantly less and thus increase cost of communication. As a future work we are tying to improve goodput over multi-hop network of TigerCENSE node.

During the time of fetching of images from the camera, Radio transciever is in Idle state, this time thus is $89.6seconds$. Total time radio-transciever is in idle state thus is $((0.01 * 86400) + 89.6) = 953.6s$. We will be sending images taken in two batches of 10 images each to the base station. We have already assumed the size of images to be 40 Kilobytes. If the Goodput is 10 Kilobytes/second. It will take us $(40 * 10)/10 = 40seconds$ to transport all the images to the base station. The total amount of time radio transciever will be on in 24 hours transporting images is $80seconds$.

These estimate of energy consuption for microcontroller and Radio-Transciever were calculated for deRFmega128 module[18].

*4) IR Flash :* There are 12 Infrared LED(s) that form part of design of IR Flash, these are used to illuminate body of tiger to allow us to take images at night. Each of these Led(s) draw 1 Ampere of current and are switched on for a period of 20ms. The total current draw for 12 LED(s) is 12 Ampere for a period of 20ms. There are twenty images taken in a day, leading to total time of operation of LED(s) of IR Flash to be $20ms * 20 = 400ms$.

### D. Images under IR Flash

We have tested this node using design of Infra-Red flash as mentioned in [4] in laboratory. Figure 6, shows image taken under room-light. Figure 5, shows image taken under Infra-Red flash. The approximate distance of the node from the whiteboard in the image was 15 feets.

## V. Conclusion

In this paper we have discussed a design of Wireless Image Sensing node that improves upon existing design of TigerCENSE, we further discuss use of ContikiOS in design of our node and use of 6LoWPAN implementation in ContikiOS to network these nodes, and results of performance evaluation of node in laboratory. We are presently working on improving goodput of the nodes in an energy efficient manner over multi-hop links, a suitable mechanism to reduce idle listening of the node and an improved design of Infrared flash based on NCP-5680, and subsequently field-testing the developed nodes.

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Fig. 5. Image taken using C328 Camera module under Infra-Red Flash in pitch darkness

Fig. 6. Image taken using C328 camera module under Roomlights

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